# CONTENT MATURITY, RELEASE LIFECYCLE & PLAYLIST ROTATION ANALYSIS

Spain Top 50 Songs

**Project Report**

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Data Analytics Project | Python · Pandas · Plotly · Streamlit

This report summarizes lifecycle and playlist-churn measures calculated from the provided processed CSV files. Results describe observed snapshots and should not be interpreted as causal explanations of song popularity.

# Executive Summary

This project analyzes 545 song lifecycle records and 554 comparisons between observed playlist snapshots from the Spain Top 50 dataset. The lifecycle analysis measures observed appearances, first and last observed dates, best recorded chart position, calendar span, and time to peak. The churn analysis measures new entries, disappearances, retained tracks, retention, and exit rates.

The average observed appearance count was 50.90 (median 13). The mean calendar span was 60.92 days (median 16 days). Average retention across the snapshot comparisons was 96.53%, with a median of 98.00%. These aggregate values describe the supplied processed data; they do not establish why a song remained on or left a playlist.

### Project Objectives

• Prepare and validate playlist observations for analysis.
• Describe song lifecycle and recorded chart performance.
• Categorize songs using project-defined lifecycle thresholds.
• Compare consecutive snapshots to quantify playlist rotation.
• Present results through a Streamlit dashboard and downloadable data.

# Data and Scope

The lifecycle file contains 545 rows and the churn file contains 554 snapshot comparisons. The lifecycle file includes song title, entry date, exit date, observation count, peak position, peak date, calendar span, time to peak, and lifecycle stage. The supplied lifecycle file does not include an artist column, so song identity is based on the records and grouping already represented in that file.

The churn comparisons span from 18 May 2024 to 27 November 2025. Snapshot intervals may vary, so each comparison represents the change between the available observations.

# Methodology

**1. Lifecycle metrics.** Observed appearances count the records in which a song was seen. Calendar span is the date difference between the first and last observed dates as provided in the processed file. Time to peak is measured from first observed date to the recorded peak date. These measures depend on the available snapshot schedule.

**2. Lifecycle classification.** The supplied file groups songs into Short-lived, Developing, and Established categories. Based on the project's stated rules, these stages are associated with appearance-count thresholds: up to 7, 8–30, and more than 30 observations respectively.

**3. Playlist churn.** New entries and disappearances are compared between successive snapshots. Retention rate is retained tracks divided by tracks in the previous snapshot, multiplied by 100. Exit rate is disappearing tracks divided by tracks in the previous snapshot, multiplied by 100.

**4. Visualization.** A Streamlit dashboard presents KPI summaries, lifecycle visualizations, searchable details, and churn charts, with CSV download functionality.

# Results: Lifecycle Analysis

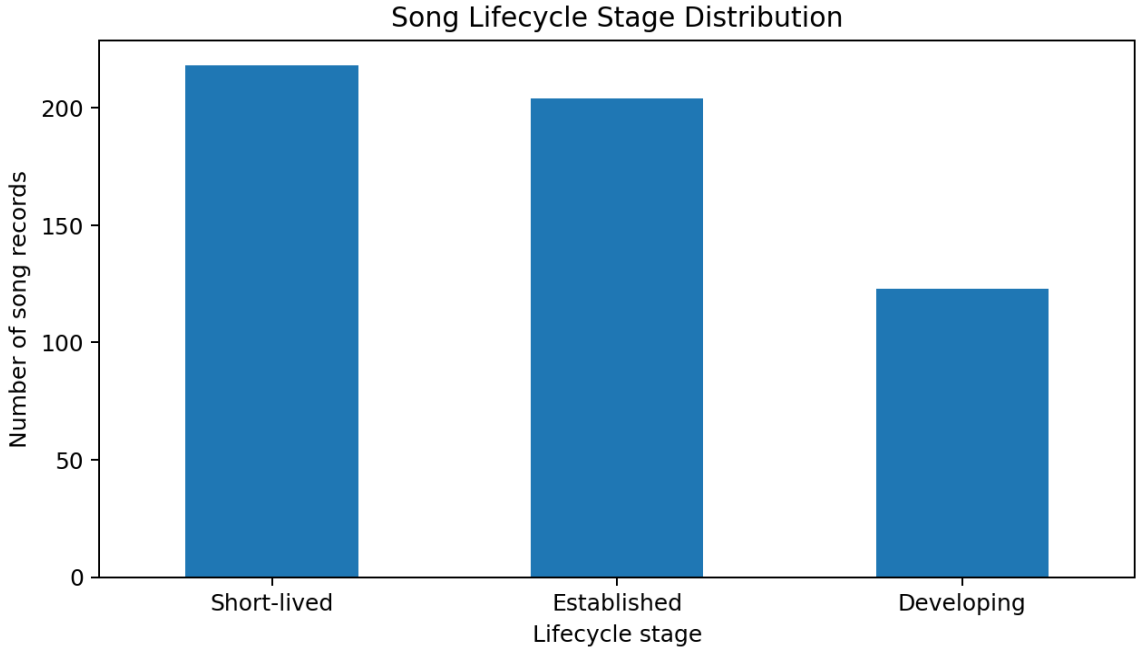
| Metric | Result |
|---|---|
| Song lifecycle records | 545 |
| Mean observed appearances | 50.90 |
| Median observed appearances | 13 |
| Mean best recorded chart position | 21.77 |
| Mean calendar span | 60.92 days |
| Median calendar span | 16 days |
| Mean time to peak | 10.31 days |
| Median time to peak | 1 days |

## Interpretation

The mean appearance count (50.90) is higher than the median (13), suggesting a right-skewed distribution in which a subset of songs has many more observed appearances than the typical record. Similarly, the mean calendar span (60.92 days) is substantially above the median (16 days), indicating variation in observed lifecycle spans. Best recorded chart position is interpreted inversely: lower values represent better positions.

## Lifecycle Stage Distribution

| Lifecycle stage | Count | Share |
|---|---|---|
| Short-lived | 218 | 40.00% |
| Established | 204 | 37.43% |
| Developing | 123 | 22.57% |




The stage labels are project-defined categories based on observed appearance counts; they are not official Spotify classifications.

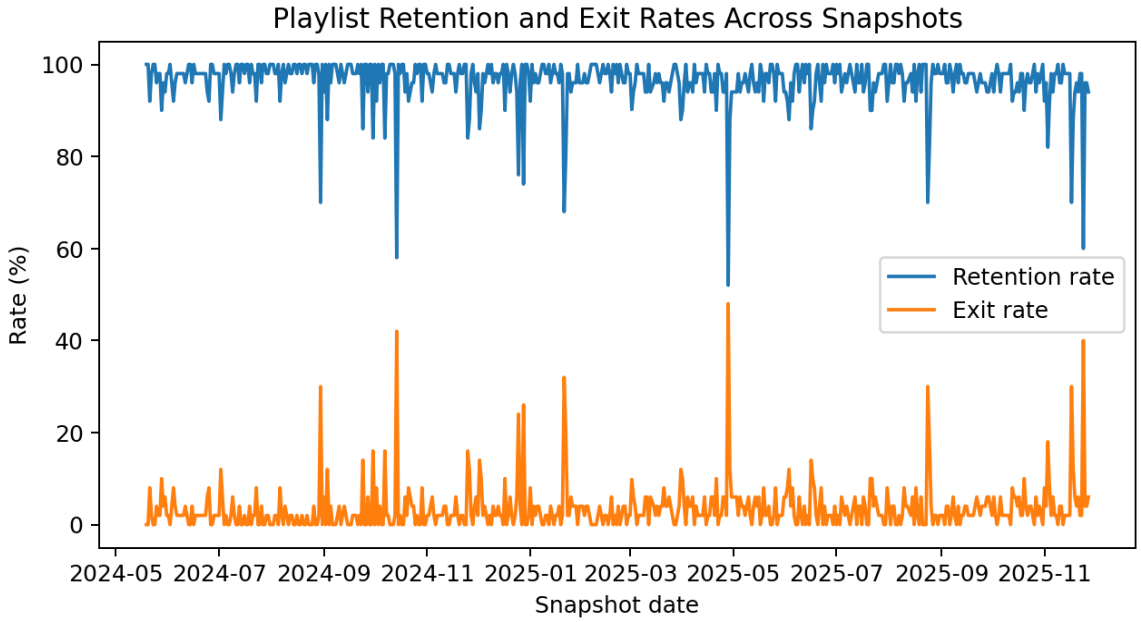
# Longest Observed Calendar Spans

| Song | First observed | Last observed | Span (days) | Appearances |
|---|---|---|---|---|
| La Bachata | 18 May 2024 | 19 Nov 2025 | 551 | 538 |
| El Merengue | 18 May 2024 | 18 Nov 2025 | 550 | 487 |
| AMARGURA | 26 May 2024 | 31 Oct 2025 | 524 | 153 |
| VAGABUNDO | 18 May 2024 | 16 Oct 2025 | 517 | 388 |
| Columbia | 08 Jul 2024 | 02 Nov 2025 | 483 | 463 |

These songs have the largest calendar spans in the supplied lifecycle file. A calendar span is the interval between the first and last observed dates; it does not prove continuous playlist presence on every intervening day.

# Results: Playlist Churn Analysis

| Metric | Result |
|---|---|
| Snapshot comparisons | 554 |
| Total new entries | 961 |
| Total disappearances | 961 |
| Mean new entries per comparison | 1.73 |
| Mean retained tracks | 48.27 |
| Mean retention rate | 96.53% |
| Median retention rate | 98.00% |
| Mean exit rate | 3.47% |
| Comparisons with no entries or disappearances | 156 |




The largest turnover comparison occurred on 28 April 2025: 24 new entries, 24 disappearances, and 52% retention.

# Discussion

The lifecycle measures show that observed song presence varies considerably. The gap between mean and median appearances suggests that a subset of records appears much more often than the typical record. Lifecycle categories make this variation easier to summarize, but their meaning depends on the chosen thresholds.

The churn measures indicate that many snapshot comparisons show high retention, while some comparisons contain substantial changes. The mean retention rate and median retention rate summarize different aspects of the distribution; the median is less influenced by unusually high- or low-retention comparisons.

These findings can inform further exploration of playlist behavior, but they do not show that lifecycle duration causes chart performance or that a particular factor caused a song to enter or leave the playlist.

# Business Value

The workflow provides a practical template for monitoring observed playlist composition and comparing song lifecycle patterns. Analysts could use similar metrics to flag large changes between snapshots, investigate songs with sustained observed presence, and guide further research into release metadata or artist-level patterns when those fields are available.

# Limitations

- The analysis is limited to the supplied processed CSV files and their upstream assumptions.
- The lifecycle file has no artist column, so records cannot be independently disambiguated by artist in this report.
- First and last observed dates may not equal actual playlist entry and removal dates.
- Calendar span does not establish continuous presence between observations.
- Churn metrics compare available snapshots; irregular intervals can affect interpretation.
- Lifecycle categories depend on the project-defined thresholds.
- Descriptive analysis does not establish causation or explain user listening behavior.

# Conclusion

The analysis combines 545 lifecycle records with 554 playlist snapshot comparisons. It quantifies observed appearances, lifecycle span, recorded peak position, and playlist turnover. The results suggest meaningful variation in song presence and generally high retention across many comparisons, alongside occasional high-turnover snapshots. These conclusions should be interpreted within the limits of the available observations.

# Recommendations for Future Work

1. Include artist identifiers and stable track IDs to distinguish songs reliably.
2. Document the original data source, collection method, and observation schedule.
3. Compare lifecycle patterns with release dates and other metadata when available.
4. Investigate unusually high-turnover snapshots and verify snapshot intervals.
5. Add automated data-quality checks and reproducible analysis steps.
6. Validate dashboard metrics against the underlying CSV outputs before publishing.

# References and Data Provenance

The calculations in this report were derived from the two user-provided processed files: *song_lifecycle_with_stages.csv* and *daily_churn.csv*. The original dataset URL, collection details, and any external literature were not included in these files, so they have not been fabricated here. Add the original source and any references actually consulted before submitting this as a formal research paper.

# Appendix: Metric Definitions

**Observed appearances:** number of recorded observations for a song.
**Calendar span:** difference between first and last observed dates.
**Best recorded chart position:** minimum numerical position recorded for a song.
**Retention rate:** retained tracks divided by previous-snapshot tracks, multiplied by 100.
**Exit rate:** disappearing tracks divided by previous-snapshot tracks, multiplied by 100.